Introduction to Probability

Department of Computer Science

August 11, 2026



Conditional Probability with tree Diagrams

Example

A bag contains 4 red balls and 5 blue balls. Mothusi picks 2 balls at random. Calculate the probability that he selects the same coloured ball each time, given that after each time a ball is selected, it is replaced.

Solution

Step 1: Define the events

- There are 4 red balls and 5 blue balls.
- Total number of balls $= 4 + 5 = 9$.
- Probability of selecting a red ball on any draw:

$$P(R) = \frac{4}{9}$$

Conditional Probability with tree Diagrams (cont.)

Solution

- Probability of selecting a blue ball on any draw:

$$P(B) = \frac{5}{9}$$

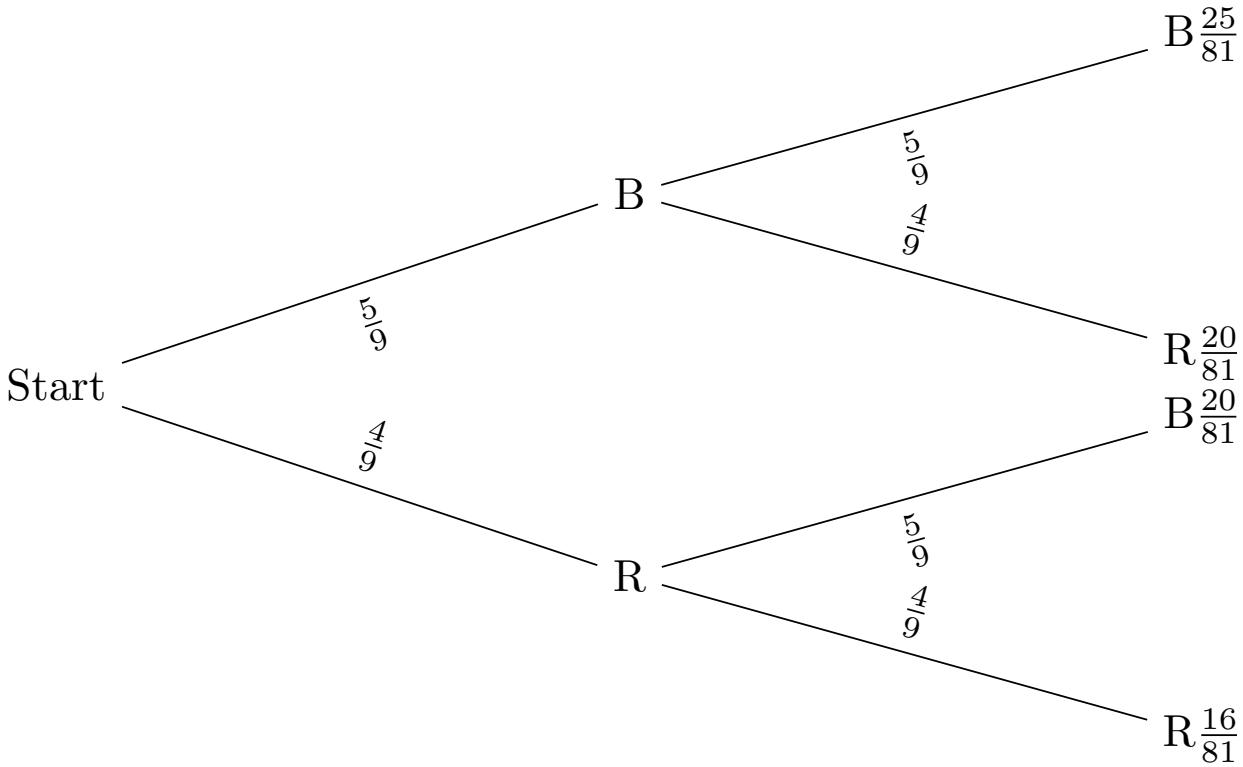
Step 2: Same colour means either (Red, Red) or (Blue, Blue)
Since the selections are with replacement, the draws are independent. Therefore:

$$P(\text{same colour}) = P(RR) + P(BB)$$

$$P(RR) = P(R) \times P(R) = \frac{4}{9} \times \frac{4}{9} = \frac{16}{81}$$

$$P(BB) = P(B) \times P(B) = \frac{5}{9} \times \frac{5}{9} = \frac{25}{81}$$

Conditional Probability with tree Diagrams (cont.)



Step 3: Add the probabilities

$$P(\text{same colour}) = \frac{16}{81} + \frac{25}{81} = \frac{41}{81}$$

Conditional Probability with tree Diagrams (Q2)

Example

: A bag contains 4 red balls and 5 blue balls. Thuma picks 2 balls at random. Calculate the probability that he selects the same coloured ball each time, given that after each time a ball is selected, it is not replaced.

Solution

Step 1: Define the events

- There are 4 red balls and 5 blue balls.
- Total number of balls $= 4 + 5 = 9$.
- The probability of selecting two balls of the same colour (without replacement) is:

$$P(\text{same colour}) = P(\text{RR}) + P(\text{BB})$$

Conditional Probability with tree Diagrams (Q2 cont.)

Solution

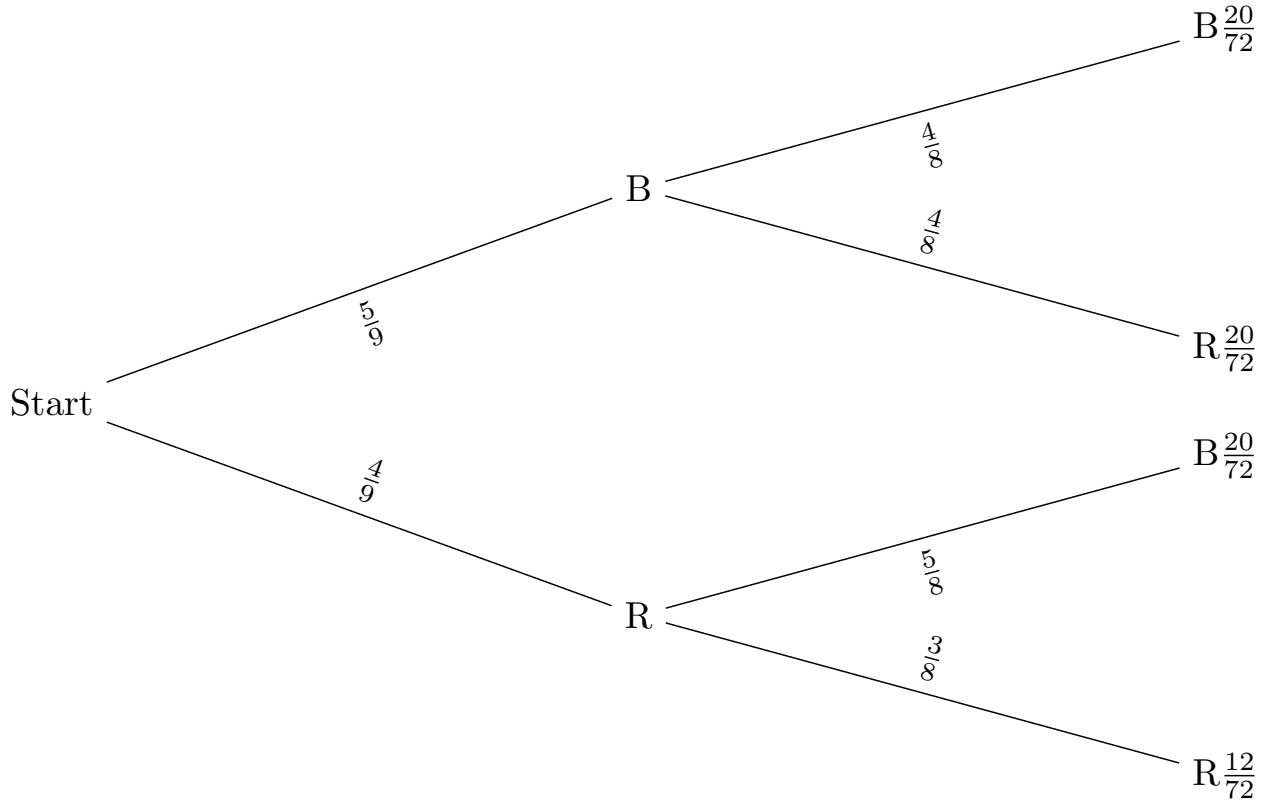
Case 1: Both balls are Red (RR)

$$P(\text{RR}) = \frac{4}{9} \times \frac{3}{8} = \frac{12}{72}$$

Case 2: Both balls are Blue (BB)

$$P(\text{BB}) = \frac{5}{9} \times \frac{4}{8} = \frac{20}{72}$$

Conditional Probability with tree Diagrams (Q2 cont.)



Final Probability: Add the probabilities

$$P(\text{same colour}) = \frac{12}{72} + \frac{20}{72} = \frac{32}{72} = \frac{4}{9}$$